Exercise Sheet 7

Selected Probabilistic Algorithms

Probabilistic Machine Learning

Prof. Dr. Michael Granitzer



7.A Mixtures and EM

Exercise 7.A.1: One EM Iteration by Hand

Four points $x = (0, 2, 3, 6)$, two Gaussian components with **fixed** $\sigma_1 = \sigma_2 = 1$, starting from $\pi = (0.5, 0.5)$ and $\mu = (1, 5)$.

1. Show that the responsibility of component 2 is $\gamma_{i2} = 1/\left(1 + e^{12 - 4x_i}\right)$ and evaluate it for the four points.
2. Do one M-step: compute N_1, N_2, the new π and the new μ_1, μ_2.
3. What does k-means (same start) do with the point $x = 3$ in its first assignment step?

Exercise 7.A.1: One EM Iteration by Hand — Solution

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1. With equal weights and variances, the normalising constants cancel: $\gamma_{i2} = \frac{e^{-(x-5)^2/2}}{e^{-(x-1)^2/2}+e^{-(x-5)^2/2}} = \frac{1}{1+e^{((x-5)^2-(x-1)^2)/2}}$ and $(x-5)^2-(x-1)^2 = 24-8x$. Values: $x=0$: $6\cdot 10^{-6}\approx 0$; $x=2$: $1/(1+e^4)\approx 0.018$; $x=3$: 0.5; $x=6$: ≈ 1.

2. $N_1 = 1+0.982+0.5+0 \approx 2.48$, $N_2 \approx 1.52$; $\pi \approx (0.62, 0.38)$. $\mu_1 = (0+2\cdot 0.982+3\cdot 0.5)/2.48 \approx 1.40$, $\mu_2 = (2\cdot 0.018+3\cdot 0.5+6)/1.52 \approx 4.96$.

3. $x=3$ is at distance 2 from both centroids: k-means must break the tie arbitrarily and gives it weight 1 in one mean and 0 in the other. EM gives it half to each — the honest answer.

Exercise 7.A.2: What EM Still Drops

1. Take a Gaussian mixture with two components and at least two distinct data points. Set $\mu_1 = x_1$ and let $\sigma_1 \to 0$, keeping component 2 fixed and wide. What happens to the likelihood — and why is that not a good model? Why does k-means not have this problem?

2. EM is run for $K = 1, \ldots, 4$. The log-likelihoods are:

K	1	2	3	4
training	−412.3	−371.8	−365.1	−360.4
held-out	−208.9	−186.5	−188.2	−191.7

   Which K do you choose, and why not the one with the best training value?

3. In which sense is EM a *partial* shortcut? Name what it conditions on the data and what it keeps as a point.

Exercise 7.A.2: What EM Still Drops — Solution

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1. The term for x_1 contains $\pi_1 \mathcal{N}(x_1 \mid x_1, \sigma_1^2) = \pi_1 / (\sqrt{2\pi}\sigma_1) \to \infty$, while every other point keeps a finite, positive density from component 2. The likelihood is unbounded — its "maximum" is a spike on one data point, which predicts nothing. k-means has no variance to shrink: its objective (SSE) is bounded below by 0. Remedies: a floor on σ_k, or a prior on it (MAP).

2. $K = 2$: the held-out log-likelihood is highest there. The training value always rises with K, because a larger mixture contains the smaller one — it is the training-error trap of Lecture 06, for densities.

3. It conditions the **hidden labels** on the data (the E-step is Bayes and keeps the whole distribution γ_{ik}). It keeps the **parameters** π, μ, σ as a single maximum-likelihood point — Lecture 03's peak — with no statement of how sure it is about them.

7.B Bayesian Regression

Exercise 7.B.1: Bayesian Regression in One Parameter

A price model **through the origin**: $y = wx + \varepsilon, \varepsilon \sim \mathcal{N}(0, 40^2)$, with x the living area in units of 100 m² and y in k€. Prior $w \sim \mathcal{N}(0, 1000^2)$. Data:

x	0.6	0.8	1.0	1.2
y	215	275	355	420

1. Compute the posterior precision $1/s_N^2 = 1/\tau^2 + \sum x_i^2/\sigma^2$, then s_N and $m_N = s_N^2 \sum x_i y_i/\sigma^2$.
2. Give the predictive mean, noise sd, parameter sd and total sd at $x = 1$ (100 m²) and at $x = 6$ (the 600 m² villa).
3. Four villas of 600 m² are added. Recompute the parameter sd and the total sd at $x = 6$ (the posterior mean hardly moves).
4. Instead, a thousand more flats of **100 m²** are added. In this model the parameter sd at 600 m² falls to about 7.6 k€. Why — and why should that make you suspicious, compared with the lecture's model with an intercept?

Exercise 7.B.1: Bayesian Regression in One Parameter — Solution

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1. $\sum x_i^2 = 3.44$, $\sum x_i y_i = 1{,}208$. Precision $= 10^{-6} + 3.44/1{,}600 \approx 0.002151$, so $s_N^2 \approx 465$, $s_N \approx 21.6$ k€ per 100 m²; $m_N \approx 465 \cdot 1{,}208/1{,}600 \approx 351$.

2. Parameter variance $x^2 s_N^2$. At $x = 1$: mean 351, noise 40, parameter 21.6, total $\sqrt{465 + 1{,}600} \approx 45$ k€. At $x = 6$: mean $\approx 2{,}106$, noise 40, parameter $6 \cdot 21.6 \approx 129$, total ≈ 135 k€.

3. $\sum x_i^2 = 3.44 + 4 \cdot 36 = 147.44$, precision ≈ 0.0922, $s_N^2 \approx 10.9$. At $x = 6$: parameter sd $\approx 6 \cdot 3.3 \approx 19.8$, total $\sqrt{391 + 1{,}600} \approx 44.6$ k€. The noise term is untouched.

4. Without an intercept, every line passes through the origin, so pinning down the slope at 100 m² pins it down everywhere: the model *assumes* that price is proportional to area all the way to 600 m². The small parameter term is bought by that assumption, not by data near the villa. With an intercept (Lecture 07), lines pivot around the data and the parameter term grows with the distance from it. A confident extrapolation is only as good as the model family — the term that no formula shows.

Exercise 7.B.2: The Two Terms in numpy

Reproduce the lecture's table ($\sigma = 40$ k€, prior $\mathcal{N}(\mathbf{0}, \mathrm{diag}(500^2, 10^2))$). Fill in the three blanks.

```
import numpy as np
x = np.array([40, 44, 48, 53, 57, 61, 65, 69, 74, 78,
              82, 86, 91, 95, 99, 103, 107, 112, 116, 120.])
y = np.array([157, 207, 114, 241, 199, 188, 200, 204, 248, 294,
              299, 314, 255, 272, 384, 379, 430, 415, 357, 443.])
sigma, S0 = 40.0, np.diag([500.0**2, 10.0**2])

def posterior(x, y):
    X = np.c_[np.ones_like(x), x]
    SN = ...                                  # (1) posterior covariance
    mN = ...                                  # (2) posterior mean
    return mN, SN

mN, SN = posterior(x, y)
for q in [80, 120, 300, 600]:
    v = np.array([1.0, q])
    param_var = ...                           # (3) parameter term
    print(q, v @ mN, sigma, np.sqrt(param_var), np.sqrt(param_var + sigma**2))
```

Then add ten villas, `x_v = np.linspace(550, 650, 10).round()` with `y_v = [1900, 1984, 2014, 2056, 2133, 2068, 2133, 2148, 2239, 2325]`, and print the row for 600 m² again.

Exercise 7.B.2: The Two Terms in numpy — Solution

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```
SN = np.linalg.inv(np.linalg.inv(S0) + X.T @ X / sigma**2)   # (1)
mN = SN @ X.T @ y / sigma**2                                 # (2)
param_var = v @ SN @ v                                       # (3)
```

Output (rounded): 80 m^2: mean 280, parameter sd 8.9, total 41.0 · 120 m^2: 420, 17.2, 43.5 · 300 m^2: 1,049, 81.1, 90.5 · 600 m^2: 2,098, 190.8, 195.0. With the ten villas, 600 m^2: mean 2,100, parameter sd 12.6, total 41.9 — the numbers of the lecture's table.

Exercise 7.B.3: Which Term?

For each situation, say whether the extra uncertainty shows up in the **noise term** σ^2, in the **parameter term** $\mathbf{x}^{*\top}\mathbf{S}_N\mathbf{x}^*$, or in **neither** — and what would reduce it.

1. Two flats with identical area sold a month apart for €310k and €370k.
2. The model is asked about a 600 m^2 villa.
3. The model is asked about a 600 m^2 villa, and villas actually sell at a luxury premium of 5,000 €/m^2.
4. The training data contain only 5 flats instead of 20, all of 40–120 m^2.
5. A lender lends up to the mean minus two sd. Why does the lecture's lender gain about €0.3M of credit on the villa from ten comparable sales, but almost nothing on an 80 m^2 flat from a thousand more flats?

Exercise 7.B.3: Which Term? — Solution

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1. **Noise** — aleatoric relative to the attributes. More flats do not help; a new attribute (condition, floor, view) might.
2. **Parameter** — far from the data the plausible lines disagree. Data near 600 m^2 reduces it.
3. **Neither** — the band is uncertainty within the linear model; a wrong family is invisible to it. Held-out data where you predict, or comparing model families, can reveal it.
4. **Parameter**, everywhere — fewer data, wider posterior. More data reduces it.
5. At the villa the total sd is 195 k€, almost all parameter term; ten villas cut it to 42 k€, so the lent amount rises from 2,098 − 390 ≈ €1.71M to 2,100 − 84 ≈ €2.02M (in k€: 1,708 and 2,016). At 80 m^2 the total sd is already 41 k€, almost all noise; more flats cannot go below 40 k€, so the loan moves by about €2k at most.